

All angles are measured in degrees.

Let

$$\beta = \angle CBH.$$

Since $BH \perp AC$,

$$\beta = 90^\circ - \angle C, \quad \angle A = 15^\circ + \beta.$$

Because ABC is acute,

$$0 < \beta < 75^\circ.$$

Put

$$\theta = \frac{\beta}{2}, \quad 0 < \theta < 37.5^\circ.$$

0.1 1. Two geometric ratios

Scale the triangle so that $BH = 1$, place $B = (0, 0)$, and put BC on the positive x -axis. Then

$$H = (\cos \beta, \sin \beta), \quad X = (-1, 0).$$

The reflection of the orthocentre across a side lies on the circumcircle. Hence H_A and H_C are respectively the reflections of H in BC and AB . Thus

$$H_A = (\cos \beta, -\sin \beta),$$

while $BH_C = 1$ and the polar angle of H_C is

$$150^\circ - \beta.$$

Set

$$K = \cot 15^\circ = 2 + \sqrt{3}.$$

Since $CH \perp AB$, the line from H to C makes angle -15° with BC . Therefore

$$C = (\cos \beta + \sin \beta K, 0).$$

Write

$$P = C + r(H - C).$$

Then $CP/CH = |r|$. Also write

$$P = X + v(H_A - X).$$

Comparing y -coordinates gives $r = -v$. Comparing x -coordinates then gives

$$r = -\frac{1 + \cos \beta + \sin \beta K}{1 + \cos \beta - \sin \beta K}.$$

Since

$$\frac{\sin \beta}{1 + \cos \beta} = \tan \theta,$$

we obtain, with $s = \sqrt{n}$,

$$s = \frac{CP}{CH} = \frac{1 + K \tan \theta}{|1 - K \tan \theta|} = \frac{\sin(15^\circ + \theta)}{|\sin(15^\circ - \theta)|}.$$

The excluded parallel case is exactly $\theta = 15^\circ$.

We next calculate AQ/H_CQ . Reflection in AB maps the line AH , which makes angle 90° with BC , to the line AH_C . Hence AH_C makes angle

$$2 \cdot 75^\circ - 90^\circ = 60^\circ$$

with BC .

The points X, H_C lie on the unit circle centred at B , at polar angles 180° and $150^\circ - 2\theta$. Indeed,

$$H_C - X = 2 \cos(75^\circ - \theta) (\cos(75^\circ - \theta), \sin(75^\circ - \theta)),$$

so XH_C makes angle $75^\circ - \theta$ with BC .

In triangle AQH_C , the sine of the angle at A is $\sin 30^\circ = 1/2$, while the sine of the angle at H_C is

$$\sin |15^\circ - \theta|.$$

Thus the sine rule gives

$$\rho := \frac{AQ}{H_CQ} = 2 \sin |15^\circ - \theta|.$$

0.1.1 Case 1: $0 < \theta < 15^\circ$

Put $d = 15^\circ - \theta$. Equations 1–2 become

$$s = \frac{\sin(30^\circ - d)}{\sin d}, \quad \rho = 2 \sin d.$$

Therefore

$$s\rho = 2 \sin(30^\circ - d) = \cos d - \sqrt{3} \sin d = \cos d - \frac{\sqrt{3}}{2} \rho.$$

Hence

$$\cos d = \rho \left(s + \frac{\sqrt{3}}{2} \right).$$

Using $\sin d = \rho/2$,

$$1 = \rho^2 \left(\frac{1}{4} + \left(s + \frac{\sqrt{3}}{2} \right)^2 \right),$$

so

$$\boxed{\rho^{-2} = n + 1 + \sqrt{3n}}.$$

0.1.2 Case 2: $15^\circ < \theta < 37.5^\circ$

Now put $d = \theta - 15^\circ$. Then

$$s = \frac{\sin(30^\circ + d)}{\sin d}, \quad \rho = 2 \sin d.$$

Thus

$$s\rho = \cos d + \frac{\sqrt{3}}{2} \rho,$$

and similarly

$$\boxed{\rho^{-2} = n + 1 - \sqrt{3n}}.$$

0.2 2. Characterizing the amazing integers

For any fixed shape of triangle, scaling multiplies both AQ and H_CQ by the same factor. Hence some scaling makes both lengths positive integers exactly when

$$\frac{AQ}{H_CQ} = \rho$$

is rational.

Suppose n is amazing. By 3 or 4,

$$n + 1 \pm \sqrt{3n} = \rho^{-2} \in \mathbb{Q}.$$

Therefore $\sqrt{3n}$ is rational. Since $3n$ is an integer, $\sqrt{3n}$ is an integer. Divisibility by 3 then gives

$$\sqrt{3n} = 3m, \quad n = 3m^2$$

for some positive integer m . Since n is odd, m is odd.

Furthermore, 3 or 4 is an integer which is the square of a rational number, and hence is a perfect square. Consequently there is a positive integer k such that either

$$k^2 = 3m^2 + 3m + 1,$$

or

$$k^2 = 3m^2 - 3m + 1.$$

Conversely, any solution of 5 produces a valid triangle from the first case: for $s = \sqrt{3}m$, choose

$$\tan \theta = \frac{s - 1}{K(s + 1)}.$$

Then $0 < \theta < 15^\circ$, and 3 gives $\rho = 1/k$. Scaling so that $AQ = 1$ gives $H_CQ = k$.

For 6, solving 1 in the second case gives

$$\tan \theta = \frac{s + 1}{K(s - 1)}.$$

When $m = 1$, this equals 1, giving $\theta = 45^\circ$, which is not an acute-triangle configuration. If $m \geq 3$, then $s = \sqrt{3}m > 5$, and

$$\tan \theta < \frac{3/2}{3} = \frac{1}{2} < \tan 30^\circ,$$

while $\tan \theta > \tan 15^\circ$. Thus $15^\circ < \theta < 30^\circ$, and the triangle is valid.

Therefore the amazing integers correspond exactly to odd m satisfying 5, or odd $m \geq 3$ satisfying 6.

0.3 3. The Pell equations

Multiplying 5 by 4 gives

$$(2k)^2 - 3(2m + 1)^2 = 1.$$

Similarly, 6 becomes

$$(2k)^2 - 3(2m - 1)^2 = 1.$$

Let

$$\alpha = 2 + \sqrt{3}, \quad \alpha^j = u_j + v_j\sqrt{3}.$$

All positive integer solutions of

$$U^2 - 3V^2 = 1$$

are given by

$$U + V\sqrt{3} = \alpha^j, \quad j \geq 1.$$

For completeness, take any positive solution $w = U + V\sqrt{3}$. Choose j such that

$$\alpha^j \leq w < \alpha^{j+1}.$$

Then $r = w\alpha^{-j} \in [1, \alpha)$, and $r = a + b\sqrt{3}$ for integers a, b , with norm 1. Since $r \geq 1$, we have $a \geq 1, b \geq 0$. If $b \geq 1$, then $a^2 = 1 + 3b^2$, so $a \geq 2$, giving

$$r = a + b\sqrt{3} \geq 2 + \sqrt{3} = \alpha,$$

a contradiction. Hence $r = 1$, proving the assertion.

Modulo 2,

$$u_{j+1} \equiv v_j, \quad v_{j+1} \equiv u_j.$$

Thus u_j is even and v_j odd exactly when j is odd.

Also, since

$$\alpha^2 = 7 + 4\sqrt{3},$$

we have

$$v_{j+2} = 4u_j + 7v_j \equiv -v_j \pmod{4}.$$

Starting from $v_1 = 1$, this gives

$$v_j \equiv \begin{cases} 1 \pmod{4}, & j \equiv 1 \pmod{4}, \\ j \equiv 3 \pmod{4}. \end{cases}$$

For equation 7, $v_j = 2m + 1 \equiv 3 \pmod{4}$, because m is odd. Hence

$$j \equiv 3 \pmod{4}, \quad m = \frac{v_j - 1}{2}.$$

For equation 8, $v_j = 2m - 1 \equiv 1 \pmod{4}$, so

$$j \equiv 1 \pmod{4}, \quad m = \frac{v_j + 1}{2}.$$

The case $j = 1$ gives $m = 1$, which was excluded. Therefore the amazing numbers correspond, in order, to every odd exponent

$$j = 3, 5, 7, 9, \dots$$

and

$$n_i = 3 \left(\frac{v_{2i+1} + (-1)^i}{2} \right)^2.$$

These are increasing because

$$v_{j+2} = 4u_j + 7v_j > v_j + 2,$$

so the correction by ± 1 cannot reverse the order.

In particular,

$$n_{2i} = 3 \left(\frac{v_{4i+1} + 1}{2} \right)^2.$$

0.4 4. Determining x and z

Since $\alpha^{-1} = 2 - \sqrt{3}$,

$$v_j = \frac{\alpha^j - \alpha^{-j}}{2\sqrt{3}}.$$

Thus 10 gives

$$n_{2i} = 3 \left(\frac{\alpha^{4i+1} - \alpha^{-(4i+1)}}{4\sqrt{3}} + \frac{1}{2} \right)^2.$$

Consequently,

$$\frac{n_{2i}}{\alpha^{8i}} \longrightarrow 3 \left(\frac{\alpha}{4\sqrt{3}} \right)^2 = \frac{\alpha^2}{16}.$$

It follows that

$$n_{2i}^{1/i} \longrightarrow \alpha^8.$$

Hence no $x > \alpha^8$ can satisfy the required eventual lower bound.

Moreover, writing $A_i = \alpha^{4i+1}$, [$\frac{A_i - A_i^{-1}}{4\sqrt{3}} + \frac{1}{2}$

$\frac{A_i}{4\sqrt{3}}$,] because $(A_i^{-1}/(4\sqrt{3})) < 1/2$. Therefore [$n_{2i} > \frac{A_i^2}{16} = \frac{\alpha^2}{16} (\alpha^8)^i$.] Thus [$x = \alpha^8$, qqquad $z = \frac{\alpha^2}{16}$.]

Finally, [$x + 2016z = \alpha^8 + 126\alpha^2$.] Now

$$\alpha^2 = 7 + 4\sqrt{3},$$

and

$$\alpha^8 = (97 + 56\sqrt{3})^2 = 18817 + 10864\sqrt{3}.$$

Hence

$$x + 2016z = (18817 + 10864\sqrt{3}) + 126(7 + 4\sqrt{3}) = 19699 + 11368\sqrt{3}.$$

Therefore

$$p + q + r = 19699 + 11368 + 3 = \boxed{31070}.$$